INCH-POUND

MIL-I-48634 (AR)
AMENDMENT 3
05 May 1993

SUPERSEDING
AMENDMENT 2
20 February 1989

MILITARY SPECIFICATION

INTEGRATED CIRCUIT, DIGITAL, CMOS LOGIC ARRAY MONOLITHIC SILICON

This Amendment forms a part of Military Specification MIL-I-48634 (AR), dated 31 July 1986, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1: Change "M718E1/M741E1" to "M718A1/M741A1".

2.1.1: Under Specification, MILITARY, Add MIL-A-48078: Ammunition, Standard Quality Assurance Provisions, General Specification For.

PAGE 3

3.2.5: Delete in its entirety and substitute: "Pin Assignments. Pin assignments shall be as shown in Figure 2.

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FIGURE 2: Delete "GD" and substitute "NC" on Pin 4.

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3.4.1: Change "^{o}C" to "^{o}F" under Unit.

HEADER: Delete "MIL-I-48643 (AR)" and substitute "MIL-I-48634 (AR)".

TABLE I: Delete "^{o}F" and substitute "^{o}C" under Unit.

3.4.3.2, third line down: Delete "V_{OD}" and substitute "V_{DD}".

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3.4.3.3, sixth line down: Delete "3 V_{DD}" and substitute "2 V_{DD}".

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TABLE II: Delete in its entirety and add new TABLE II included.

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* 3.3.3, Delete paragraph in its entirety.

PAGE 41

3.5.2, last sentence: Delete "3.5.15" and substitute "3.4.3.3".

PAGE 42

3.5.5, second line down: Delete "15 cycles" and substitute "15 cycles".

Delete "3.5.15" and substitute "3.4.3.3".

3.5.6, last sentence: Delete "3.5.15" and substitute "3.4.3.3".

PAGE 43

4.3.5, last sentence: Delete "(see paragraph 5.3)" and substitute "(see paragraph 5.3.1)".

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Add the following paragraph:

"4.6 Screening. Screening shall be in accordance with MIL-STD-883, method 5004 with the exception of the following:

Screen	Method
Burn-In Test	1015 at 125°C for 160 hours
Post Burn-In	1015 at 25°C all electrical parameter per Table II.

* 4.4.2, last line, Delete "design activity" and substitute "technical agency".

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TABLE II. PERFORMANCE CHARACTERISTICS: T_A = 25 °C

PARAMETER	TEST CONDITIONS	OUTPUT PINS NO.	OUTPUT CLOCK CYCLE	MIN	MAX	UNITS
V_{OH}	V_{DD} = 5.0V I_{OH} = 0.1 mA Connect a 46.4K resistor to V_{SS}	1, 2, 9, 17 10 11 8 3	88466 0 2051 2052 Note 1	4.6	N/A	VOLTS
	V_{DD} = 10.0 V I_{OH} = 0.1 mA Connect a 95.3K resistor to V_{SS}	1, 2, 9, 17 10 11 8 3	88466 0 2051 2052 Note 1	9.5	N/A	VOLTS
V_{OL}	V_{DD} = 5.0 V I_{OL} = 0.2 mA Connect a 23.7 K resistor to V_{DD}	8, 9, 11 10 3	0 88466 Note 1		0.40	VOLTS
	V_{DD} = 10.0 V I_{OL} = 0.2 mA Connect a 47.5 K resistor to V_{DD}	8, 9, 11 10 3	0 88466 Note 1		0.50	VOLTS
	V_{DD} = 5.0 V I_{OL} = 0.4 mA Connect a 11.5 K resistor to V_{DD}	1, 2, 17	0		0.40	VOLTS
	V_{DD} = 10.0 V I_{OL} = 0.4 mA Connect a 23.7 K resistor to V_{DD}	1, 2, 17	0		0.50	VOLTS

NOTE 1: This output (PTB) is not reset by T/O (Pine 18). PTB stays high for 64 clock cycles and low for 320 clock cycles and may be measured at any clock count which achieves the desired state.

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TABLE II. (CONT'D)

PARAMETER	TEST CONDITIONS	OUTPUT PINS NO.	MIN	MAX	UNITS
V_{IH} VOLTS	V_{DD} = 0.5 V	6, 7, 16, 18	3.5		
	V_{DD} = 10.0 V		7.0		
V_{IL}	V_{DD} = 5.0 V	6, 7, 16, 18		1.5	VOLTS
	V_{DD} = 10.0 V			3.0	
I_{IH}	V_{IN} = 5.0 V V_{DD} = 5.0 V	6, 7, 16, 18		0.1	uA
	V_{IN} = 10.0 V V_{DD} = 10.0 V			0.2	
I_{IL}	V_{IN} = 0.2 V V_{DD} = 5.0 V	6, 7, 16, 18	-0.1		uA
	V_{IN} = 0.5 V V_{DD} = 10.0 V		-0.2		
I_{DD}	V_{DD} = 7.0 V	5 STATIC All inputs @ V_{IL}, outputs unloaded		10	uA
		DYNAMIC Input Pattern of Table III @ 1 KHz Clock Rate		100	uA
T_{PD} and other timing parameters	Compliance with paragraph 3.4 @ V_{DD} = 5.0 Volts and CLK = 1.0 KHz (pin 7) shall constitute verification that internal propagation delay and timing parameters satisfy the requirements of this specification.				

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Add the following figure:

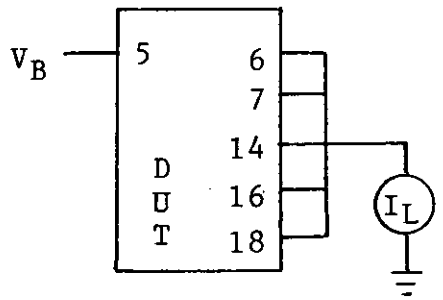


FIGURE 3. LEAKAGE CURRENT TEST.

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* 6.4, Delete "ammunitions data cards" in 2 places.

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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